www.PapaCambridge.com

O

ADDITIONAL MATHEMATICS **4037/21**

Paper 2 **May/June 2010**

2 hours

Additional Materials: Answer Booklet/Paper

7629663991

This document consists of **6** printed pages and **2** blank pages.

DCA (KN) 12931/4
© UCLES 2010

UNIVERSITY of CAMBRIDGE
International Examinations

[Turn over

www.PapaCambridge.com

Mathematical Formulae

1. ALGEBRA

Quadratic Equation

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Binomial Theorem

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \ldots + \binom{n}{r} a^{n-r} b^r + \ldots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{(n-r)!r!}$.

2. TRIGONOMETRY

Identities

$$\sin^2 A + \cos^2 A = 1.$$

$$\sec^2 A = 1 + \tan^2 A.$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A.$$

Formulae for ΔABC

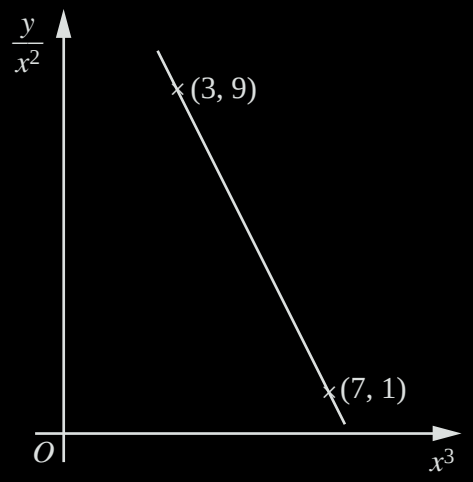
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

$$a^2 = b^2 + c^2 - 2bc \cos A.$$

$$\Delta = \frac{1}{2} bc \sin A.$$

www.PapaCambridge.com

1



The variables x and y are related so that, when $\frac{y}{x^2}$ is plotted against x^3, a straight line graph passing through (3, 9) and (7, 1) is obtained. Express y in terms of x. [4]

2 In a singing competition there are 8 contestants. Each contestant sings in the first round of this competition.

(i) In how many different orders could the contestants sing? [1]

After the first round 5 contestants are chosen.

(ii) In how many different ways can these 5 contestants be chosen? [2]

These 5 contestants sing again and then First, Second and Third prizes are awarded to three of them.

(iii) In how many different ways can the prizes be awarded? [2]

3 It is given that $x - 1$ is a factor of f(x), where $\text{f}(x) = x^3 - 6x^2 + ax + b$.

(i) Express b in terms of a. [2]

(ii) Show that the remainder when f(x) is divided by $x - 3$ is twice the remainder when f(x) is divided by $x - 2$. [4]

www.PapaCambridge.com

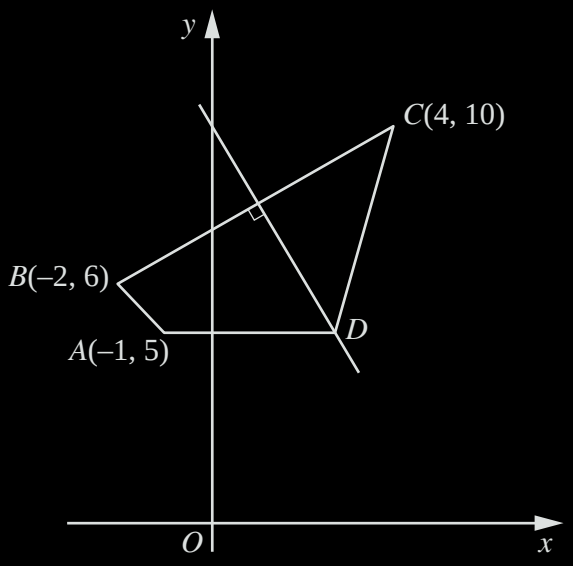
4 **(a)** Given that $\sin x = p$ and $\cos x = 2p$, where x is acute, find the exact value of p and the value of $\operatorname{cosec} x$.

(b) Prove that $(\cot x + \tan x)(\cot x - \tan x) = \dfrac{1}{\sin^2 x} - \dfrac{1}{\cos^2 x}$.

5 Given that a curve has equation $y = x^2 + 64\sqrt{x}$, find the coordinates of the point on the curve where $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$. [7]

6 The line $y = x + 4$ intersects the curve $2x^2 + 3xy - y^2 + 1 = 0$ at the points A and B. Find the length of the line AB. [7]

7 **Solutions to this question by accurate drawing will not be accepted.**



In the diagram the points $A(-1, 5)$, $B(-2, 6)$, $C(4, 10)$ and D are the vertices of a quadrilateral in which AD is parallel to the x-axis. The perpendicular bisector of BC passes through D. Find the area of the quadrilateral $ABCD$. [8]

www.PapaCambridge.com

8 **(a)** Given that $\mathbf{A} = \begin{pmatrix} 2 & 3 & 7 \\ 1 & -5 & 4 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 2 & 1 \\ 8 & 6 \end{pmatrix}$, calculate

(i) $2\mathbf{A}$,

(ii) $\mathbf{B}^2$, [2]

(iii) $\mathbf{BA}$. [2]

(b) **(i)** Given that $\mathbf{C} = \begin{pmatrix} 2 & 1 \\ 7 & 6 \end{pmatrix}$, find $\mathbf{C}^{-1}$. [2]

(ii) Given also that $\mathbf{D} = \begin{pmatrix} 4 & 3 \\ -2 & -1 \end{pmatrix}$, find the matrix $\mathbf{X}$ such that $\mathbf{XC} = \mathbf{D}$. [2]

9 A particle starts from rest and moves in a straight line so that, t seconds after leaving a fixed point O, its velocity, v ms^{-1}, is given by

$$v = 4\sin 2t.$$

(i) Find the distance travelled by the particle before it first comes to instantaneous rest. [5]

(ii) Find the acceleration of the particle when $t = 3$. [3]

10 In this question, $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is a unit vector due east and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ is a unit vector due north.

A lighthouse has position vector $\begin{pmatrix} 27 \\ 48 \end{pmatrix}$ km relative to an origin O. A boat moves in such a way that its position vector is given by $\begin{pmatrix} 4 + 8t \\ 12 + 6t \end{pmatrix}$ km, where t is the time, in hours, after 1200.

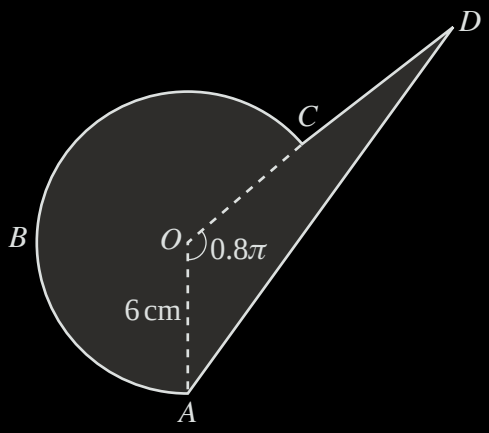
(i) Show that at 1400 the boat is 25 km from the lighthouse. [4]

(ii) Find the length of time for which the boat is less than 25 km from the lighthouse. [4]

www.PapaCambridge.com

11 Answer only **one** of the following two alternatives.

EITHER

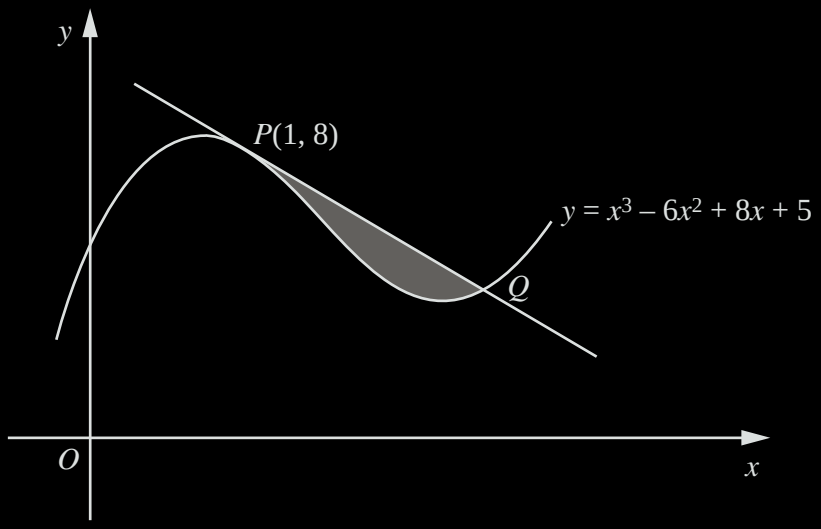


The diagram represents a company logo *ABCDA*, consisting of a sector *OABCO* of a circle, centre *O* and radius 6 cm, and a triangle *AOD*. Angle $AOC = 0.8\pi$ radians and *C* is the mid-point of *OD*. Find

(i) the perimeter of the logo, [7]

(ii) the area of the logo. [5]

OR



The diagram shows part of the curve $y = x^3 - 6x^2 + 8x + 5$. The tangent to the curve at the point $P(1, 8)$ cuts the curve at the point *Q*.

(i) Show that the *x*-coordinate of *Q* is 4. [6]

(ii) Find the area of the shaded region. [6]

BLANK PAGE

www.PapaCambridge.com

www.PapaCambridge.com

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

University of Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.